



OFFSHARE

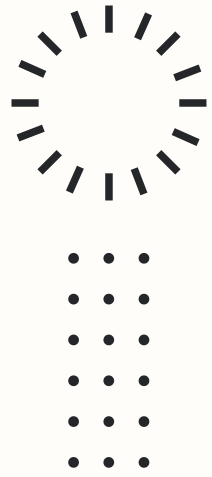


Enabling seamless, efficient, and reliable printing, offline.

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PROBLEM STATEMENT

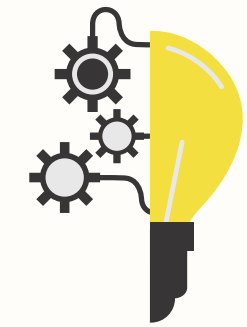
- Manual xerox shop operations lead to long queues, delays, and inefficient document submission.
- Billing and order tracking are prone to errors due to manual calculations.
- Many small xerox shops lack digital transaction records and organized print queues.
- Existing systems depend entirely on internet connectivity, which is often unreliable in campuses or local areas.



PROPOSED SOLUTION

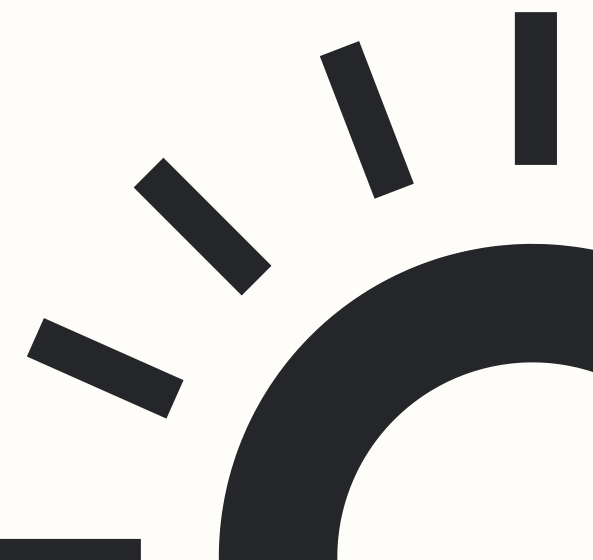
- Offshare Xerox – a hybrid online + offline xerox management system.
- Offline Mode: Customers scan a QR code to connect to the shop's local WiFi and upload documents directly.
- Online Mode: Users can select a shop from the app and send documents remotely through the internet.
- System provides automatic page detection, cost calculation, and secure transaction storage for efficient shop management.





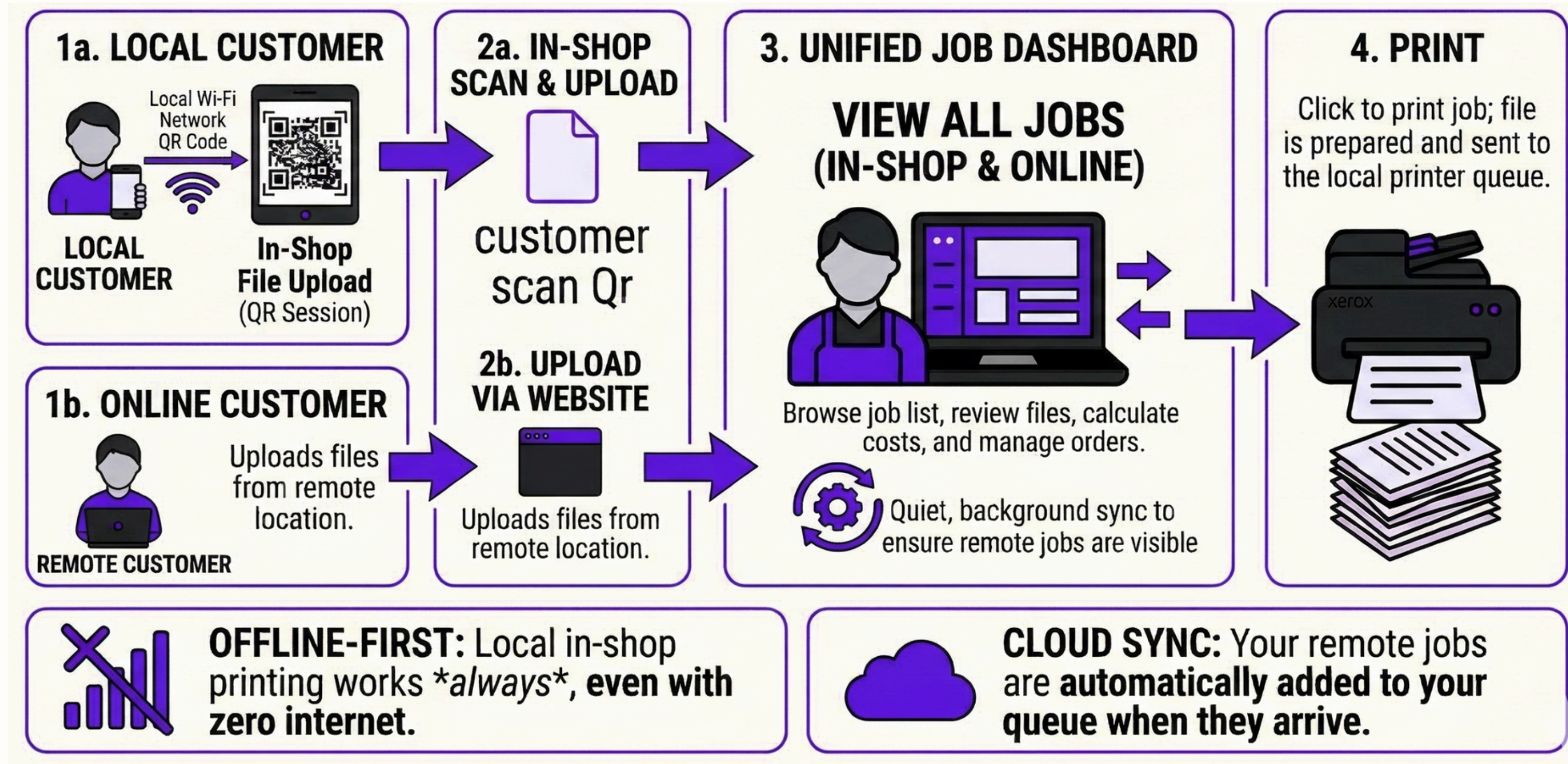
OBJECTIVES:-

- To provide a hybrid printing system supporting both online and offline document submission.
- To enable fast document transfer through QR-based local Wi-Fi connectivity inside the shop.
- To automate page detection and cost calculation for accurate and quick billing.
- To maintain secure digital transaction records with local storage and optional cloud synchronization.
- To help shopkeepers manage print queues, orders, and customer sessions efficiently.





Workflow



Entities:

Sr.No	Table	Description
1	users	Customers using the platform (online or via QR offline)
2	shops	Print shop registrations
3	shop_address	1:1 with shops — stores location + local hotspot IP for QR
4	network_config	Shop Wi-Fi/hotspot details used to generate QR codes
5	printers	Physical printers registered under a shop
6	pricing	Per-shop pricing rules (color, paper size, finishing)
7	storage	File storage references (Cloundinary, S3, local, etc.)
8	sessions	One per user visit — online or offline (QR-initiated)
9	files	Uploaded documents (PDF, DOCX, JPG, PNG, TXT)
10	print_settings	Reusable print config (color mode, sides, copies, paper)
11	print_jobs	Core job entity — the heart of the system
12	print_job_pricing	Junction table — links each job to its applied pricing rule(s)
13	payments	Payment records (Cash, UPI, Card) with Razorpay fields
14	notifications	Alerts for customers (job done) and shopkeepers (new job)
15	activity_log	Full audit trail (upload, print, payment, login, QR scan, etc.)
16	outbox	Sync queue for offline → online reconciliation (outbox pattern)
17	qr_tokens	One-time tokens embedded in QR codes for offline sessions



First Normal Form (1NF)

Rule	How Satisfied
Atomic values only	All columns store single values. JSON capabilities field is intentionally flexible but internally structured.
No repeating groups	No arrays or multi-valued attributes in any column.
Unique rows	Every table has a surrogate INT primary key.
Consistent data types	ENUM columns enforce domain consistency instead of free VARCHAR.

Second Normal Form (2NF)

Table	Partial Dependency Removed	How
print_settings	Settings lived inside print_jobs	Moved to own table; job references settings_id
shop_address	Address fields in shops	1:1 decomposition; address → shop_id, not shop attributes
storage	Storage path was in files	Storage is its own entity; multiple files can share storage
pricing	Pricing embedded in print_jobs	Pricing is a shop-level rule, not a job property

Third Normal Form (3NF)

Table	Transitive Dependency Eliminated
shops → shop_address	city/pincode/area depend on location, not on shop_name or status
print_jobs → payments	Payment details don't depend on print job they depend on the transaction
files → storage	Path/checksum describe storage, not the document itself
print_jobs → print_settings	color_mode/copies/sides describe a config profile, not a specific job
sessions → activity_log	Event descriptions depend on the event, not on session metadata

Anomaly Prevention

Anomaly Type	Example	How Schema Prevents It
Insert Anomaly	Cannot add a shop without a printer	shops table is independent of printers
Insert Anomaly	Cannot log payment without a user	payments → print_jobs → sessions → users (chain, not direct)
Delete Anomaly	Deleting a job removes file metadata	ON DELETE RESTRICT prevents cascading data loss
Delete Anomaly	Deleting a user wipes sessions/logs	ON DELETE CASCADE is used only for child-specific data
Update Anomaly	Printer IP must be updated in many rows	Printer data lives once in printers table, referenced by FK
Update Anomaly	Pricing rule changed, must update all jobs	Jobs reference pricing_id; one update propagates everywhere



Impact:-

- Improves xerox shop workflow and queue management
- Reduces billing and manual calculation errors
- Saves time for students and staff
- Decreases rush and crowding at shops



BENEFITS:-

- Supports both online and offline printing
- Fast file transfer via QR-based local Wi-Fi
- Automatic page counting and cost calculation
- Secure digital transaction records



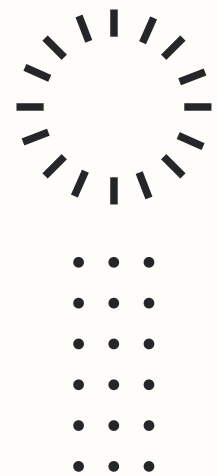
Future Scope:

- Integration of UPI / QR-based payments
- Real-time order tracking and queue status
- Support for multiple shops in one platform
- Role-based access for admin and staff
- Batch printing for multiple files



Research and References:

- Research Paper 1 : Overview of Android Application Wireless Filesharing Tool:Wi-fi-Direct based Ad Hoc Network Technology
- Research Paper 2 : A Comparative Study of Different File Sharing Applications and Wi-Fi Direct Technology for File Sharing
- Best File Sharing Apps without Internet: Share Files Offline



THANK YOU..

